Foundations of Computer Science

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Introductory

Greek letters

- α (alpha)
- β (beta)
- ɣ (gamma)
- δ (delta)
- ϵ (epsilon)
- μ (mu)
- σ (sigma)
- φ (phi)
- Ψ (psi)
- ω (omega)

Mathematical properties

- Associative: a + (b + c) = (a + b) + c
- Commutative: a + b = b + a
- Transitive: A > B, B > C, then A > C
- Distributivity: A(B + C) = AB + AC

Logic symbols

- ∀ (for all)
- ∃ (there exists)
- ∄ (there does not exist)
- ∴ (therefore)
- | (such that)
- . (such that)
- : (such that)

Set theory

Basics

Definition

- **Set**: a group or collection of elements
    - Repetition does not matter
    - The elements define the set

- Two sets are the same if they both have the exact same elements.

Notation

- {0, 1} is the set with elements 0 and 1
- {0, 1} and {1, 0} are the same sets
- {0, 1, 0} and {1, 0} are the same sets
- {0,...,9} and {0,1,2,3,4,5,6,7,8,9} are the same sets
- {1,3,5,...} is the set of odd numbers

Membership

- If element x is in X, then we write $x \in X$
- If element y is not in X, we write $y \notin X$

Cardinality

- |X| is the number of elements in the set X

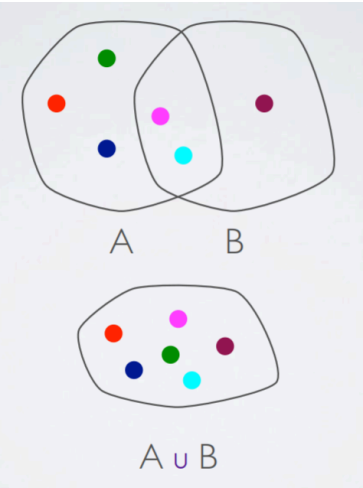
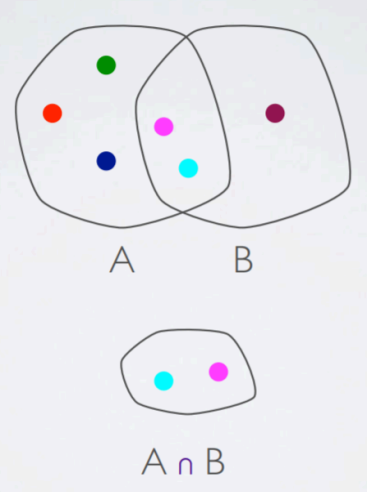
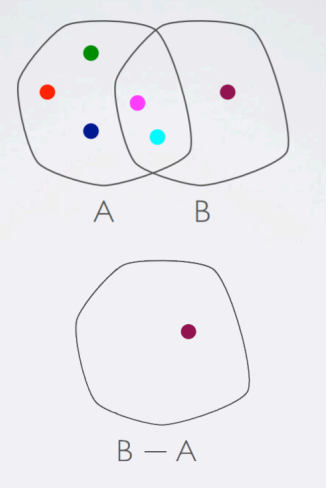
Common sets in maths

- **N** is the set of natural numbers (0, 1, 2, 3, 4...)
- **Z** is the set of integers (... -1, 0, 8, 67, 33, 20...)
- **R** is the set of real numbers (7, 23, 6/7, pi)
- **C** is the set of complex numbers (3, 9i, 6 + 2i, 9 + 12i...)

Empty set

- $\varnothing$ is the empty set. It is a set with no elements.
- $|\varnothing| = 0$

Operations on sets

Union	Intersection	Difference
		
Associative Commutative	Associative Commutative	NOT associative NOT commutative

Predicates

- **Predicate**: a function φ(x) that returns a true/false value
- When φ(x) = true, we say '*it holds for x*' or it '*satisfies x*'

Naive set theory

- You can make a subset using a predicate:
  - $\{ x \mid x \in X \ \& \ \varphi(x) \}$
  - Looks through each element of X. If φ(x) is true, x goes in the subset. If not, x is not in the subset.

Reasoning about sets

How to relate sets

- A set X is a subset of Y when every element X is also in the set Y
- $X \subseteq Y$ means X is a subset of Y
- $X \subset Y$ means X is a subset of Y, but X and Y aren't the same (proper subset)
- This is always true: $X \subseteq X$
- This is always true: $\varnothing \subseteq X$

When are two sets equal

- X and Y are equal sets when both conditions are true:
  - $X \subseteq Y$
  - $Y \subseteq X$

Products and sums of sets

Ordered pairs

- Like a set, but order matters. (0, 1) is an ordered pair.
- (0, 1) != (1, 0)

Cartesian product

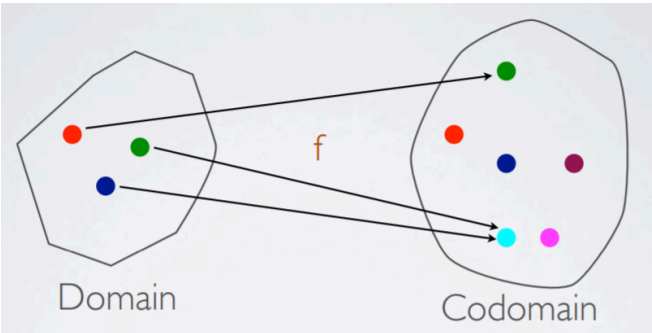
- $X \times Y = \{ (x,y) \mid x \in X \ \& \ y \in Y \}$
- $\{a, b\} \times \{x, y\} = \{ (a, x), (a, y), (b, x), (b, y) \}$
- $|X \times Y| = |X| \times |Y|$
- NOT commutative, NOT associative

Sum (or disjoint union)

- $X+Y = \{ (x,0) \mid x \in X\} \cup \{(y,1) \mid y \in Y \}$
- $\{a, b\} + \{x, y\} = \{ (a, 0), (b, 0), (x, 1), (y, 1) \}$
- $|X + Y| = |X| + |Y|$

Functions between sets

Picture of a function



Notation

- Function f that takes in elements from set X and outputs elements from set Y:
  - $f : X \rightarrow Y$
  - X is the domain
  - Y is the codomain

- Two functions are the same if, for all inputs, their outputs are the same: $f(x) = g(x)$

Range

- **Range**: Smallest subset of the codomain such that:
  - $\square x \in$ domain. $f(x) \in R$ (where R is the range)
- Every element of R can be obtained by passing through some value of x through f(x)

Functions of several variables

- If a function has multiple inputs, its domain is the cartesian product of sets
- $+ : N \times N \rightarrow N$

Operations on functions

Composition

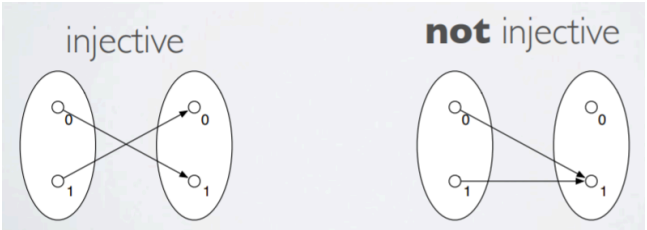
- $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ then $f ; g : X \rightarrow Z$ or $g \circ f$
- $f ; g : X \rightarrow Z = g(f(x))$
- This is associative

Inverting functions

- $f : X \rightarrow Y$ then $f^{-1} : Y \rightarrow X$ only if f has an inverse
- If $f(1) = 2$, then $f^{-1}(2) = 1$

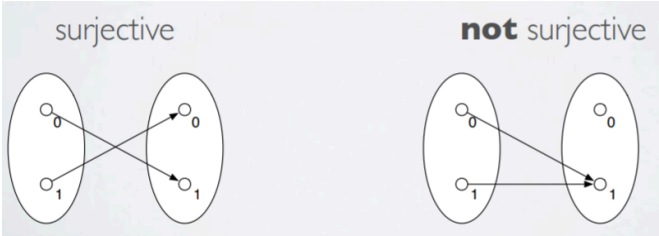
A taxonomy of functions and the isomorphism theorem

Injective functions

- A function is injective if $f(x) = f(x')$ implies that $x = x'$
- Each input must have one unique output
- 

Surjective functions

- A function is surjective if for all $y \in Y$ there exists an $x \in X$ such that $f(x)=y$
- The function must map onto the whole codomain for it to be surjective

- 

Bijective functions

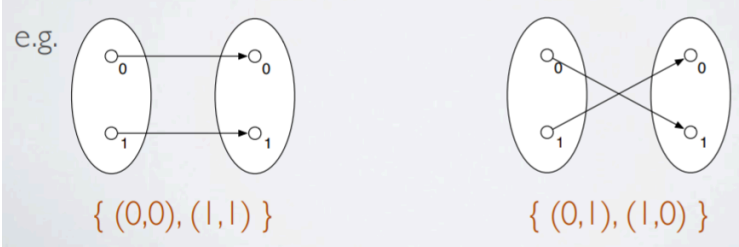
- A function is bijective if it is injective and surjective

Isomorphism theorem

- If a function is a bijection, it has an inverse.
- If a function has an inverse, it is a bijection.

Function spaces and powersets

The graph of a function

- A set can define a function like this: { (x, f(x)) | x ∈ X }
- 

Function spaces

- A function space is a set of all functions from one set to another.
- Y^X is a function space. It is a set that contains all functions from X to Y.
- $|Y^X| = |Y|^{|X|}$

Powersets

- A powerset P(X) is a set containing all the subsets of X
- P({0,1}) = { ∅, {0}, {1}, {0,1} }
- P(∅) = {∅}
- P(P(∅)) = P({∅}) = { ∅, {∅} }
- $| P(X) | = 2^{|X|}$

Sizes of infinity

Countable sets

- A set X is countable if you can make a bijection from the natural numbers (0, 1, 2, 3, 4, 5...) to the elements of X.
- {1, 2, 3, 4...} is a countable set
- **Z** = {... -2, -1, 0, 1, 2...} is a countable set
- **N x N** is a countable set

Uncountable sets

- A set X is uncountable if there are no bijections from the natural numbers (0, 1, 2, 3, 4, 5...) to the elements of X
- **R** is an uncountable set

Diagonalisation proof

- The diagonalisation proof is a type of proof that proves a set is uncountable. It works like this:
    - You want to prove set X is uncountable.
    - Set up a bijection f : **N** → X where **N** is the set of natural numbers.
    - Set up a list or a grid of possible solutions.
    - Select diagonally from that list/grid to make a new element of X.
    - Change every value of that diagonal element.
    - Show that 'f' cannot contain that new element.
    - Proof by contradiction; there is no bijection 'f'.
    - If there is no bijection, then X cannot be countable.
- Example:
- Suppose that we have a bijection f : **N**→**R**
- Where f(x) = xa,b xc,b xd,e .... where "x something, something" is a digit within the real number output of f(x).
- f(0) = x0,0 x0,1 x0,2 x0,3 x0,4 x0,5 x0,6...
- f(1) = x1,0 x1,1 x1,2 x1,3 x1,4 x1,5 x1,6...
- f(2) = x2,0 x2,1 x2,2 x2,3 x2,4 x2,5 x2,6...
- f(3) = x3,0 x3,1 x3,2 x3,3 x3,4 x3,5 x3,6...

- f(0) = **x0,0** x0,1 x0,2 x0,3 x0,4 x0,5 x0,6...
- f(1) = x1,0 **x1,1** x1,2 x1,3 x1,4 x1,5 x1,6...
- f(2) = x2,0 x2,1 **x2,2** x2,3 x2,4 x2,5 x2,6...
- f(3) = x3,0 x3,1 x3,2 **x3,3** x3,4 x3,5 x3,6...

- So now we've picked out the 'diagonal number' represented above:
- x0,0 x1,1 x2,2 x3,3 x4,4 x5,5 x6,6 ...
- Now, let's take each digit and change it's value so that each value is now different from what it was previously:

- x0,0 x1,1 x2,2 x3,3 x4,4 x5,5 x6,6 ...
- Now we have created a new number. There should be some f(x) that equals this number, because f is supposed to be a bijection.
- Is it f(0)? It can't be, because x0,0 is different from x0,0.
- Is it f(1)? It can't be, because x1,1 is different from x1,1.
- Is it f(2)? It can't be, because x2,2, is different from x2,2.
- We could keep going forever, but we'll never find an f(x) that equals this number because this number will always be different somewhere down the line.
- Therefore the function 'f' is not surjective. If it is not surjective, it cannot be bijective. If there is no bijection to map **N** to **R**, then **R** is uncountable.

Cantor's Theorem

- For any set X, the cardinality of P(X) is strictly greater.

Cantor-Bernstein-Schröder Theorem

- If there exist injective functions f: X→Y and g: Y→ X then X and Y have the same cardinality.

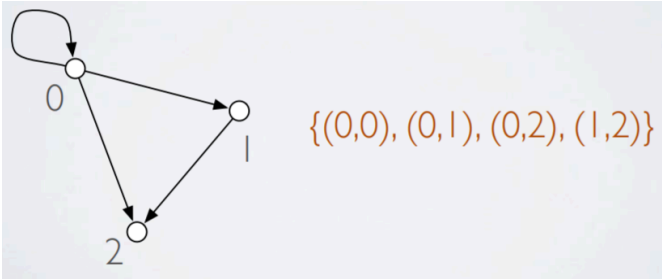
Relations

Relations between sets

- A relation between sets X and Y is the subset of X × Y
- It's like a function, but without the rules
- If R is a relation from X to X, it's a '*relation on X*'
- The full relation is X × Y
- The empty relation is ∅
- The identity relation on X is { (x, x) | x ∈ X }
- They can be composed just like functions, like this: R ; S
- A function is a special kind of relation, except that you cannot have multiple outputs with functions
- The number of possible relations from X to Y is $2^{|X||Y|}$ because relations are subsets of the cartesian product X × Y, so all possible relations between X and Y can be found in the powerset of X × Y.

Digraphs

- A digraph is a directed graph
- It can be represented as a set of ordered pairs

- 

Equivalency

Equivalence relations

- An equivalence relation is a relation ~ on a set X that satisfies 3 axioms:
    - Reflexivity: x~x
    - Symmetry: x~y, therefore y~x
    - Transitivity: x~y and y~z, therefore x~z
- An equivalence relation returns either true (they are equivalent) or false (they are not equivalent)
- An example is the equals sign **=** on the set of integers.

Equivalence classes

- An equivalence class of a is a set of all elements in X that are equivalent to a.
- It is written like this: [a]
- X/~ denotes the set of all equivalence classes with respect to ~ upon X.
- Another name for X/~ is the '*quotient of X with respect to ~*'

Partitions

- A collection of subsets of X that, when all unioned, will make up X without any overlaps.
- The set of equivalence classes X/~ is a partition of X.
- The set of the sets of all equivalence classes X/~ for every ~ on X contains all partitions of X.

Partial Orders

- A partial order is a relation ~ on the set X if it satisfies these axioms:
    - Reflexive
    - Transitive
    - Anti-symmetric: if x~y and y~x then x = y

Total Orders

- A total order is a partial order that satisfies this property:
    - For all x and y, either x~y is true or y~x is true
- An example of a total order is less than or equal to

Propositional and predicate logic

Propositional logic

Syntax

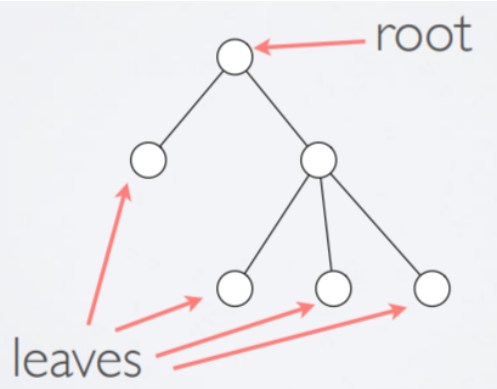
Formulas

- Formulas are built up from:
  - Propositional variable symbols (p, q, r...)
  - Unary operation symbol ¬ called negation
  - Binary operation symbol ∧, ∨, → called conjunction, disjunction and implication
  - constant symbol ⊥ called falsity, or contradiction
  - brackets (,)

Backus-Naur form

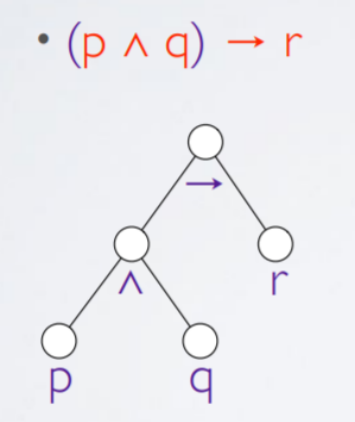
- Backus-Naur form is a notation technique to describe the syntax of languages
- The BNF of propositional logic could be this:
  - V ::= p | q | r | s | ... (propositional variables)
  - F ::= V | ⊥ | ¬F | F ∧ F | F ∨ F | F → F (formulas)
- The left-hand side must be replaced with one of the right-hand side expressions encapsulated by |. For example, if we had
  - F
- We must replace it with ‘V’, ‘⊥’, ‘¬F’ etc. In this example, we could change it to:
  - F ∧ F
- But now we must replace the other F’s, because the left-hand side is nonterminal, meaning we cannot just leave ‘F’ or ‘V’ in our formula. We keep replacing the nonterminals until we get just expressions found on the right-hand side (called terminals)
  - F ∧ F (same as before)
  - ¬F ∧ F (replace first F with ¬F)
  - ¬V ∧ V (replace both F’s with V’s)
  - ¬p ∧ q (replace both V’s with propositional variables)
- By using BNF, any formula can be made. This can be visualised as a recursive process.

Trees

- 

- A tree is a mathematical structure that has **nodes** and **arcs**.
    - The beginning node (usually at the top) is called the **root**.
    - Each node can have zero or more **children**.
    - Each node that is not the root will have a parent.
    - You cannot have any cycles.

Syntax trees

- A syntax tree is a representation of a formula using a tree.
- 

- This is sometimes called ‘abstract syntax trees’

Syntactic conventions and brackets

- ¬ binds tighter than ∧ and ∨
- ∧ and ∨ bind tighter than →
- ∧ and ∨ have the same precedence and associate to the left
- → associates to the right

Semantics

What is the semantics of a formula?

- Semantics means the ‘meaning’ of a formula: what does a formula actually mean?
    - A variable can be true (T) or false (F)

- A formula will be T or F depending on its sub-formulas

Truth tables

- A truth table is simply a grid displaying every combination of T and F for a formula's variables. You can show semantics using truth tables by splitting up a formula into its sub-formulas and calculating the values one by one.
-

φ	ψ	$\varphi \wedge \psi$	$\varphi \vee \psi$	$\varphi \rightarrow \psi$
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Semantics as a (recursively defined) function

- **2** = { F, T }
- $\neg : \mathbf{2} \rightarrow \mathbf{2}$
- $\vee : \mathbf{2} \times \mathbf{2} \rightarrow \mathbf{2}$
- $\wedge : \mathbf{2} \times \mathbf{2} \rightarrow \mathbf{2}$
- $\rightarrow : \mathbf{2} \times \mathbf{2} \rightarrow \mathbf{2}$
- **V** = { p, q, r...} (set of propositional variables)
- σ: **V** $\rightarrow$ **2** (sets values to the variables, e.g. $\sigma(p)$ = T will set variable p to true)
- $[\![$ some formula $]\!]_\sigma$: **F** $\rightarrow$ **2** (this function will take a formula and convert it to a truth value through recursion, by splitting up a function into its subfunctions, a.k.a using semantics)
- Using this function, you can do:
  - $[\![\bot]\!]_\sigma = F$
  - $[\![p]\!]_\sigma = \sigma(p)$
  - $[\![\varphi \vee \psi]\!]_\sigma = \vee([\![\varphi]\!]_\sigma, [\![\psi]\!]_\sigma)$
  - $[\![\varphi \wedge \psi]\!]_\sigma = \wedge([\![\varphi]\!]_\sigma, [\![\psi]\!]_\sigma)$
  - $[\![\varphi \rightarrow \psi]\!]_\sigma = \rightarrow([\![\varphi]\!]_\sigma, [\![\psi]\!]_\sigma)$
  - $[\![\neg \varphi]\!]_\sigma = \neg([\![\varphi]\!]_\sigma)$
- As you can see, this function will split up a formula into its sub-formulas until it reaches the variables, at which it will replace it with its truth value in the σ function.

Tautologies

- If a formula is T no matter what truth values its variables are, it's called a **tautology**.
- If a formula is T through at least one assignment of its variables, it's **satisfiable**.
- Two formulas are logically equivalent when they evaluate to equal truth values for any assignment of truth values of all their variables
  - Note: logical equivalence is an equivalence relation on the set of all formulas!

Formal proofs

Semantic entailment

- If a formula φ is a tautology (always true no matter what), write ⊨ φ
- The symbol ⊨ is called "semantic entailment"
- Suppose that Γ is a set of formulas
- Γ ⊨ φ (read Γ semantically entails φ) means: if all of the formulas in Γ are true, then φ will be true.
- ⊨ φ is a shorthand for ∅ ⊨ φ because you don't need other formulas to be true for φ to be true; φ will always be true no matter what because it's a tautology.

Formal proof systems

- A formal proof system is a collection of rules that you can use to prove theorems.
- Natural deduction is an example of a formal proof system.
- Γ ⊢ φ means that we can use the formulas in Γ in a formal proof system to prove φ
- If you can prove φ without using any other formulas, write ⊢ φ which is a shorthand for ∅ ⊢ φ.

Soundness

- 'Soundness' is a property of a proof system.
- If Γ ⊨ φ is true, then the proof system with Γ is sound.
- Soundness basically means that if you manually plugged in true and false values into Γ and φ, or if you used a truth table to prove it, it would all work fine. It doesn't necessarily mean that you can prove it, but you still know that it's correct.
- A way to visualise soundness is this:
    - You have this mathematical formula:
    - $sin^2(x) + cos^2(x) = 1$
    - And you want to prove it to others. However, you don't really know how to prove it; you just know that it works. You could punch a bunch of values in your calculator to show that it works, but you still haven't proved it for all cases.
    - This is an example of sound (because it's right), but not complete (because you can't prove it).

Completeness

- 'Completeness' is a property of a proof system.
- If Γ ⊢ φ is true, then the proof system with Γ is complete.
- Completeness basically means that you can use the proof system with Γ to prove that φ is true. It doesn't necessarily mean that it's correct, it just means that you can prove it.
- A way to visualise completeness is this:
    - You start with the following formula
    - $a = b$

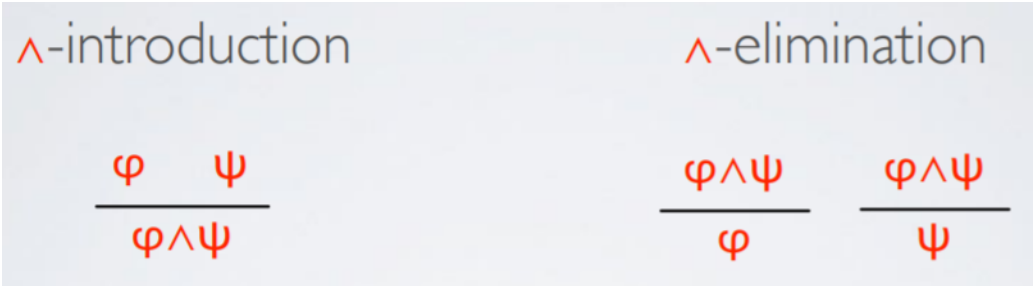
- Multiply by a on both sides
- $a^2 = ba$
- Subtract b squared
- $a^2 - b^2 = ba - b^2$
- Factorise both sides
- $(a - b)(a + b) = b(a - b)$
- Divide out (a - b)
- $(a + b) = b$
- Substitute in a for b because a = b
- $(b + b) = b$
- Combine terms
- $2b = b$
- Divide by b
- $2 = 1$
- This proof is not correct, because it allows division by zero (a - b is zero).
- This is an example of complete (because you've proved it) but not sound (because you're not correct).

Natural deduction

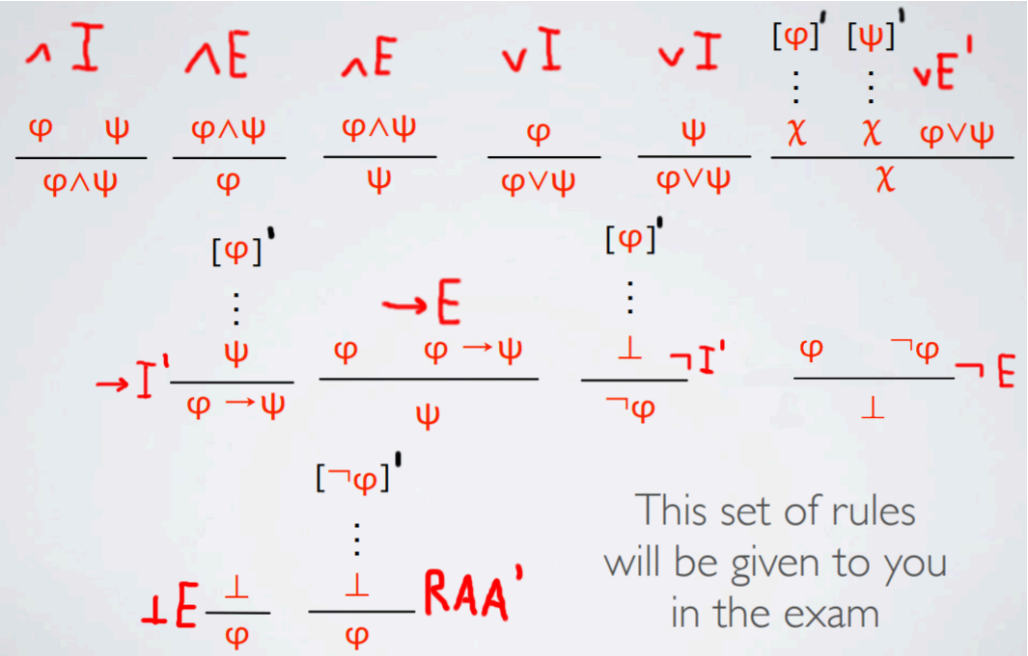
Proofs in natural deduction

- A proof can be seen as a tree with:
- leaves as assumptions
- the root as the proved formula (Q.E.D.)
- the nodes are determined by proof rules
- Proof rules fall into two categories:
- Introduction - introduces a logical connective
- Elimination - eliminates a logical connective

Rules of natural deduction

Rules for conjunction	

Rules for implication	$\rightarrow$-introduction $$\frac{\begin{array}{c}[\varphi]\\ \vdots \\ \psi\end{array}}{\varphi \rightarrow \psi}$$ $\rightarrow$-elimination (modus ponens) $$\frac{\varphi \qquad \varphi \rightarrow \psi}{\psi}$$
Discharging assumptions	$\wedge$-introduction $$\frac{\varphi \qquad \psi}{\varphi \wedge \psi}$$ $\rightarrow$-introduction $$\frac{\begin{array}{c}[\varphi]\\ \vdots \\ \psi\end{array}}{\varphi \rightarrow \psi}$$ $$\frac{\dfrac{\dfrac{[\varphi]^1 \quad [\psi]^2}{\varphi \wedge \psi}\wedge I}{\varphi \rightarrow \varphi \wedge \psi}\rightarrow I^1}{\psi \rightarrow (\varphi \rightarrow \varphi \wedge \psi)}\rightarrow I^2$$
Rules for negation	$\neg$-introduction $$\frac{\begin{array}{c}[\varphi]\\ \vdots \\ \bot\end{array}}{\neg \varphi}$$ $\neg$-elimination $$\frac{\varphi \qquad \neg \varphi}{\bot}$$

Rules for disjunction	
All	

Properties of natural deduction

- Natural deduction is sound and complete.

Predicate logic

What's the difference?

- Propositional logic talked about true and false values
- Predicate logic talks about elements of a set X
- A predicate is a function that returns true or false.

Arities

- Given a set X, a function with arity $n \in \mathbf{N}$ (also sometimes called n-ary function) on X is a function with type $X^n \to X$
- It's basically a function that takes in n parameters and outputs a single output

- Examples are nullary (n = 0), unary (n = 1), binary (n = 2), ternary (n = 3) etc.

Example predicate logic formulas

- Example: Let friends be a binary relation symbol, where friends(x,y) means that y is a friend of x.
  - $\forall$x. $\forall$y. friends(x,y) → friends(y,x)
  - If x is a friend of y, y is a friend of x
  - $\forall$x. $\exists$y. friends(x,y)
  - Everyone has at least one friend
  - $\exists$x. $\forall$y. friends(x,y)
  - There exists at least one person who is friends with everyone

Syntax

BNF for predicate logic

- Variables:
  - V
- Terms:
  - T ::= V | f(T, T,...,T)
  - f(T, T,...,T) means any function capable of n T's as parameters
  - Terms must be treated as sets and variables of sets
- Formulas:
  - F ::= R(T, T,...,T) | ⊥ | ¬F | F ∧ F | F ∨ F | F → F | $\exists$x. F | $\forall$x. F
  - R(T, T,...,T) means any relation capable of n T's as parameters
  - Formulas 'F' must be treated like propositional variables because they will always return true or false.

Free and bound variables

- A variable in a formula is either free or bound.
  - Free is when a variable could be anything
  - Bound is when a variable is in the scope of a quantifier (like $\exists$ there exists or $\forall$ for each)
- Example:
  - f(x) = y (x, y free)
  - $\forall$x. f(x)=y (x bound, y free)
  - $\exists$y. $\forall$x. f(x) = y (x, y bound)

Syntactic conventions

- The scope of quantifiers exceeds all and covers all
  - $\forall$x,y. R(x,y) → S(y,x)
  - is
  - $\forall$x,y. (R(x,y) → S(y,x))
- Common abbreviation for quantifier repetition:
  - $\forall$x,y. φ(x,y)

- is
- $\forall x.\ \forall y.\ \varphi(x,y)$

Semantics

Semantics of predicate logic

- A set A called the domain
  - For each n-ary function symbol f, a function $f : A^n \rightarrow A$
  - For each n-ary relation symbol R, a subset $R \subseteq A^n$
- Given a function σ from variables to elements of A, any term now has a meaning as an element of A

Semantics of terms

- $[\![x]\!]_\sigma = \sigma(x)$
- $[\![f(t1,t2,...,tk)]\!]_\sigma = f([\![t1]\!]_\sigma,[\![t2]\!]_\sigma,...,[\![tk]\!]_\sigma)$

Linear algebra

Vectors

- A vector is a column of numbers that maps one position to another on a space
- An example of a vector: $\begin{bmatrix} a \\ b \end{bmatrix}$

Gaussian Elimination

Row reduction

- If we had:
  $3x + 2y + z = 39$
  $2x + 3y + z = 34$
- $x + 2y + 3z = 26$
- To reduce this, we could:
  - Add row 1 to −3 times row 3, and set that to row 3
  - Add row 1 to $(-\frac{3}{2})$ times row 2, and set that to row 2
  - Add $(-\frac{8}{5})$ times row 2 to row 3, and set that to row 3
- This will be three transformations:
- $$\xrightarrow[\rho 3]{\rho_1 - 3\rho_3} \circ \xrightarrow[\rho 2]{\rho_1 - \frac{3}{2}\rho_2} \circ \xrightarrow[\rho 3]{-\frac{8}{5}\rho_2 + \rho_3}$$
- And will transform the set of equations into this (row echelon form):
- $$\begin{aligned} 3x + 2y + z &= 39 \\ -\frac{5}{2}y - \frac{1}{2}z &= -12 \\ -36z &= -99 \end{aligned}$$
- From there, we can make all the first coefficients 1:
- $$\begin{aligned} 1x + \frac{2}{3}y + \frac{1}{3}z &= 13 \\ 1y - \frac{1}{5}z &= \frac{24}{5} \\ 1z &= \frac{11}{4} \end{aligned}$$
- With this, you can solve it by substitution.

Matrix representations

- You can represent that previous problem using matrices in the form **A**x = b:
- $$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 39 \\ 34 \\ 26 \end{bmatrix}$$
- You could add syntactic sugar to this by using an augmented matrix (**A**|b):
- $$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array}\right]$$
- You can perform all of the operations as mentioned previously to achieve this augmented matrix:
- $$\left[\begin{array}{ccc|c} 1 & \frac{2}{3} & \frac{1}{3} & 13 \\ 0 & 1 & -\frac{1}{5} & \frac{24}{5} \\ 0 & 0 & 1 & \frac{11}{4} \end{array}\right]$$
- Reduced row echelon form is when you have row echelon form, but all the 1's are the only non-zero elements in their columns, e.g:
- $$\left[\begin{array}{cccc|c} 1 & 0 & 4 & 0 & 4 \\ 0 & 1 & 7 & 0 & 7 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right]$$

Row operations

- Row operations:
  - Swap 2 rows: $\rho_i \leftrightarrow \rho_j$
  - Multiply all entries in a row by a non-zero constant: $\rho_i \leftarrow k\rho_i$
  - Replace a row by the sum of itself and a multiple of another row: $\rho_i \leftarrow \rho_i + k\rho_j$
- Two matrices are row equivalent if you can derive one from the other using these operations.
- If two augmented matrices are row equivalent, they will have the same solutions.

How many solutions

- Get row echelon form:
  - If there is a contradiction (e.g. 0 = 1), **no solutions**
  - If there are no contradictions and each row has a pivot, **unique solution**
  - If there are no contradictions, but there is at least 1 row with no pivot, **infinite solutions**

Row operations as matrix multiplication

- Let:
- $$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$
- Swap two rows:
- $$P_{23} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}$$
- Multiply all entries in a row by a non-zero constant
- $$(I + (x-1)E_{22})A = \begin{bmatrix} a & b & c \\ xd & xe & xf \\ g & h & i \end{bmatrix}$$
- Replace a row by the sum of itself and a multiple of another row
- $$(I + xE_{32})A = \begin{bmatrix} a & b & c \\ d & e & f \\ xd+g & xe+h & xf+i \end{bmatrix}$$
- The definitions of P and E are as follows:
- $$(\mathbf{E}_{rs})_{ij} = \delta_{ri}\delta_{sj} \text{ where } \delta_{pq} = \begin{cases} 1 & p = q \\ 0 & \text{otherwise.} \end{cases}$$
- $$(\mathbf{P}_{rs})_{ij} = \begin{cases} 1 & i = j \text{ unless } i = j = r, \text{ or } i = j = s \\ 1 & (i,j) = (r,s) \text{ or } (i,j) = (s,r) \\ 0 & \text{otherwise.} \end{cases}$$

Linear dependence

- **A**<u>x</u> = <u>b</u> where <u>b</u> = 0 is a homogeneous equation
- <u>x</u> = 0 is always a solution
- <u>x</u> != 0 is a non-trivial solution

Finding linear dependence with algebra

- If you can find
- $$x \begin{bmatrix} a \\ b \\ c \end{bmatrix} + y \begin{bmatrix} d \\ e \\ f \end{bmatrix} + z \begin{bmatrix} g \\ h \\ i \end{bmatrix} = 0$$
  $$x, y, z \neq 0$$
- Those vectors are **linearly dependent**

- If you cannot find a non-trivial solution, then those vectors are **linearly independent**

Finding linear dependence with calculus

- Let's just say you had vectors and sets of vectors:
- $$V_1 = \begin{bmatrix} 4 \\ -2 \end{bmatrix} \quad V_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad V_3 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$ $$S_1 = \{V_1, V_2\} \quad S_2 = \{V_1, V_3\}$$
- Since:
- $$\frac{1}{2}V_1 = V_2$$
- S_1 is **linearly dependent**.
- However, since there is no way to get V_1 from V_3, S_2 is **linearly independent**.
- If you can derive one vector from others in a set of vectors, that set of vectors is linearly dependent. If they're all completely different, then that set is linearly independent.

Kernel / Nullspace

- The kernel or nullspace of a matrix A is the set of all vectors V such that a vector v in V will map the matrix A to zero.
- Basically if you had a vector 'v' and a matrix 'A', and you multiplied them together like this: Av, but then the answer you get is 0, then the vector 'v' will be in the 'kernel' of A.
- $$ker(A) = \{v \in V \mid Av = 0\}$$
- Where V is the set of all vectors

Span

- W is in the span of V = { V_1, V_2, V_3 ... } if you can use the vectors in V to get to W.
- E.G. if W = $3V_1 + 2V_2 - 4V_3$ then W is in the span of V.
- If there are more vectors to dimensions in V, then V is **linearly dependent** and **underdetermined** (fat):
- $$V = \left\{ \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix}, \begin{bmatrix} e \\ f \end{bmatrix}, \begin{bmatrix} g \\ h \end{bmatrix} \right\}$$
- $$M = \begin{bmatrix} a & c & e & g \\ b & d & f & h \end{bmatrix}$$ (column vectors)
- If there are more dimensions to vectors, then V is **overdetermined** (thin):
- $$V = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}, \begin{bmatrix} e \\ f \\ g \\ h \end{bmatrix} \right\}$$

- $$M = \begin{bmatrix} a & e \\ b & f \\ c & g \\ d & h \end{bmatrix}$$

Matrix multiplication rules

- A(xB + yC) = xAB + yAC distributivity
- A(BC) = (AB)C associativity
- AB ≠ BA non-commutative multiplication

Transpose

- $$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$
- $$(AB)^T = B^T A^T$$
- $$(a, b)^T = \begin{bmatrix} a \\ b \end{bmatrix}$$

Inverse properties

- If A^{-1} exists, Ax = 0 admits only trivial solution.
- x |→ Ax is injective.
- Upon row reduction, every row has a pivot if inverse exists. In reduced row echelon, A is equivalent to the identity matrix.
- Columns of matrix where inverse exists are linearly independent.

Matrix determinants

Minor

- A minor $m(a_{ij})$ of matrix A is the determinant of the submatrix formed by deleting the i-th row and j-column from A:
- $$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$
  $$m(A_{21}) = det(\begin{bmatrix} b & c \\ h & i \end{bmatrix})$$

- The determinant of a 2x2 matrix, like in the above image, is:
- $$bi - ch$$

Cofactor

- The cofactor is defined with the following equation:
- $$cof(A_{ij}) = (-1)^{i+j} \times m(A_{ij})$$
- It basically adds a 1 or -1 coefficient to the minor

Determinant

- The determinant is defined with the following equation:
- $$\begin{aligned} &det(A) = \\ &A_{i1} \times cof(A_{i1}) + \\ &A_{i2} \times cof(A_{i2}) + \ldots \\ &A_{in} \times cof(A_{in}) + \end{aligned}$$
- Where i can be any row of matrix A and n is the number of columns matrix A has

Inverse of a 3x3 matrix

- Let's just say we have a matrix A:
- $$A = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$
- Firstly, we need to get its determinant.
- I will pick row 3 to calculate the determinant on, but you can pick any row, like I've mentioned above.
- To get the determinant, you'll need to calculate a row's 3 minors:
- $$A = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$
- $m(A_{31})$ = (2)(1) - (2)(1) = 0
- $$A = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$
- $m(A_{32})$ = (0)(1) - (2)(1) = -2

- $$A = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$
- $m(A_{33}) = (0)(1) - (2)(1) = -2$
- Now you will need to get their cofactors:
- $cof(A_{31}) = (-1)^{3+1} * 0 = 0$
- $cof(A_{32}) = (-1)^{3+2} * -2 = 2$
- $cof(A_{33}) = (-1)^{3+3} * -2 = -2$
- And now you can get the determinant by doing this:
- det(A) =

  $A_{31} * 0 +$
  $A_{32} * 2 +$
  $A_{33} * -2$
- det(A) =

  $0 * 0 +$
  $1 * 2 +$
  $2 * -2$
- det(A) = -2
- Hold onto that value for a moment; we'll need it later.
- Now, let's transpose matrix A. That means flipping the rows and the columns (picture it like a y = -x reflection):
- $$A = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$
- Now, find all the minors of the transposed matrix (sounds like a lot of work, but it's not that hard. It's just that there's a lot of them):

$$A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$(1)(2) - (1)(1) = \mathbf{1}$ $\qquad$ $(2)(2) - (2)(1) = \mathbf{2}$ $\qquad$ $(2)(1) - (2)(1) = \mathbf{0}$

$$A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$(1)(2) - (1)(0) = \mathbf{2}$ $\qquad$ $(0)(2) - (0)(2) = \mathbf{0}$ $\qquad$ $(0)(1) - (2)(1) = \mathbf{-2}$

$$A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$(1)(1) - (1)(0) = \mathbf{1}$ $\qquad$ $(0)(1) - (2)(0) = \mathbf{0}$ $\qquad$ $(0)(1) - (2)(1) = \mathbf{-2}$

- Now put all the minors into a matrix of their own. This is called the 'adjugate' matrix, represented by Adj(A):
- $$Adj(A) = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & -2 \\ 1 & 0 & -2 \end{bmatrix}$$
- Find the cofactors of those minors and replace the elements with them (basically add -1 and 1 coefficients). Instead of using the formula, you can just remember this matrix layout. It might be easier to remember that:
- $$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & -2 \\ 1 & 0 & -2 \end{bmatrix} \times \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$
$$= \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$
- Now times that matrix by 1 / det(A) to get the inverse of A:
- $$A^{-1} = \frac{1}{det(A)} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$
- $$A^{-1} = \frac{1}{-2} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$
- $$A^{-1} = \frac{1}{2} \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -2 \\ -1 & 0 & 2 \end{bmatrix}$$
- You have now calculated the inverse of a 3x3 matrix!

Eigenvalues and eigenvectors

- Av = λv
- λ = eigenvalue
- v = eigenvector
- If you apply the matrix transformation A on an eigenvector v, it's the same as a scalar multiplication of scale factor λ (eigenvalue).

Calculating eigenvalues

- To calculate the eigenvalues of A, set up the equation:
- $$A - \lambda I$$

- For this example, let:
- $$A = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$$
- Now, simplify that equation into one big matrix:
- $$\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
- $$\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$
- $$\begin{bmatrix} 4-\lambda & 5 \\ 3 & 2-\lambda \end{bmatrix}$$
- Find the determinant of that matrix:
- $$\begin{aligned} (4-\lambda)(2-\lambda) - (5)(3) \\ (\lambda^2 - 6\lambda + 8) - 15 \\ \lambda^2 - 6\lambda - 7 \end{aligned}$$
- Make it equal to 0, factorise and solve:
- $$\begin{aligned} \lambda^2 - 6\lambda - 7 = 0 \\ (\lambda - 7)(\lambda + 1) = 0 \\ \lambda = 7, \lambda = -1 \end{aligned}$$
- You have now found the eigenvalues of matrix A!

Calculating eigenvectors

- To calculate an eigenvector, you need its corresponding eigenvalue.
- Set up the following equation:
- $$(A - \lambda I)v = 0$$
- Where v is going to be the eigenvector you will solve for. Now, sub in A and the eigenvalue and solve for v.
- $$\begin{aligned} (\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix})v = 0 \\ (\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix})v = 0 \\ \begin{bmatrix} -3 & 5 \\ 3 & -5 \end{bmatrix} v = 0 \end{aligned}$$
- Use an arbitrary vector for v, with the dimensions making sense, so that you can solve for it.

$$\begin{bmatrix} -3 & 5 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$R_2 = R_1 + R_2 \rightarrow \begin{bmatrix} -3 & 5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-3x + 5y = 0$$

$$3x = 5y$$

$$x = \frac{5}{3}y$$

$$Let\ z = y$$

$$v = \begin{bmatrix} \frac{5}{3}z \\ z \end{bmatrix}$$

- $where\ z \in \mathbb{R}$
- Do the same thing for the other eigenvalue, and you will have calculated all the eigenvectors!

Combinatorics

Principle of inclusion and exclusion

- $|A \cup B| = |A| + |B| - |A \cap B|$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Pigeonhole principle

- If there are more elements than there are slots, then at least 2 elements will share the same slot.
- e.g:
  - You have 11 pigeons and 10 pigeonholes.
  - You don't have enough pigeonholes to allocate each pigeon a hole.
  - Therefore at least 2 pigeons must share 1 hole.

Fibonacci sequence and recursion

- The fibonacci sequence is a sequence where an element is the sum of the previous two elements
- { 1, 1, 2, 3, 5, 8, 13 ... }
- You can use recursion to define this:

- Recurrence relation: $f_n = f_{n-1} + f_{n-2}$

$$\begin{pmatrix} f_n \\ f_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} f_{n-1} \\ f_{n-2} \end{pmatrix}$$

Proof by induction

- You have a proposition P upon x
- Induction:
  - 1) Prove that P(1) is true
  - 2) Let P(n) be true
  - 3) Prove that P(n+1) is true if P(n) is true
  - 4) If P(n+1) is true, and P(1) is true, then P(2) is true.
  - 5) If P(n+1) is true, and P(2) is true, then P(3) is true.
  - 6) If P(n+1) is true, and P(3) is true, then P(4) is true etc. etc.
- Therefore proved for all Natural numbers

Counting by bijections

- If |A| is known and f : A --> B a bijection, then |B| = |A|

Bit strings

- A bit string is a way of representing one of the possible combinations of choice.
- Example:
  - Let's just say you had strawberry doughnuts, chocolate doughnuts and vanilla doughnuts.
  - A customer has to pick up to 6 doughnuts for one box. They can pick any flavours they want, as long as it's 6 doughnuts.
  - They could pick 2 strawberry, 3 chocolate and 1 vanilla.
  - You could represent this with the stars and bars method:
  - ★★|★★★|★
  - With the number of stars on the left being strawberry,
  - the number of stars in the middle being chocolate,
  - and the number of stars on the right being vanilla.
  - Bit strings are basically stars and bars, but with 0 instead of stars and 1 instead of bars:
  - 00100010
  - If you need n doughnuts, and have k types, your bit string will be of length:
    - n + k - 1
  - And the number of 1's in your bit string will be:
    - k - 1
  - The number of ways of choosing n doughnuts of k types is:

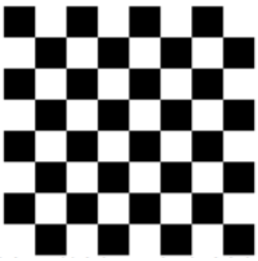
- $$\binom{n+k-1}{n}$$

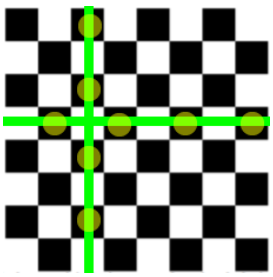
Generalised sum and product rule

Sum rule

- Use a sum + to represent ‘OR’
- Example:
    - Q. What’s the number of ways you can roll two equal even numbers with two dice?
    - You can roll two 2’s **or** two 4’s **or** two 6’s.
    - 1 + 1 + 1 = 3
    - There are 3 ways to roll two equal even numbers with two dice.

Product rule

- Use multiply * to represent ‘AND’
- Example:
    - Q. How many ways can you roll an even number and an odd number with two dice?
    - You can roll an even number 3 ways (2, 4, 6) **and** you can roll an odd number 3 ways (1, 3, 5).
    - 3 * 3 = 9
    - There are 9 ways you can roll an even number and an odd number with two dice.
- Another example:
    - In how many ways can you choose 1 white and 1 black square from 8-by-8 chessboard? If the squares cannot be on the same row or column?
    - 
    - There are 32 different white squares to pick from first.
    - Once you pick a white square, there are 8 other black squares on the same row or column.

- 
- Therefore, the second black square you pick has 8 less options.
- 32 * (32 - 8) = 768
- There are 768 different ways to choose 1 white and 1 black square.

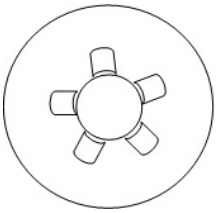
Generalised product rule

- Let's just say you had a sequence:
- { x_1, x_2, x_3, x_4 ... x_n }
- If x_1 had i_1 ways of being filled in,
- and x_2 had i_2 ways of being filled in,
- and x_3 had i_3 ways of being filled in ...
- and x_n had i_n ways of being filled in, then the number of possible combinations is:
- $i_1 * i_2 * i_3 * ... * i_n$

Permutations

- Let's just say you had to put 10 boxes into 10 slots.
- You have 10 choices to put the first box in. You put the first box into slot 1.
- Now, you have to put the second box somewhere. You can't put it in slot 1, because it's taken. Therefore, you have 9 choices to put the second box in. You put it in slot 2.
- Now, you have to put the third box somewhere. You can't put it in slots 1 or 2, because they're both taken. Therefore, you have 8 choices to put the second box in.
- Following this through, you find out you have 10 * 9 * 8 * 7 * ... * 2 * 1 combinations. The shorter way to write this is 10! (factorial). 10! = 3628800.

Circular arrangements

- Let's just say you have a table with 5 seats:
- 
- And you want to arrange people in it. How many different ways can you arrange everyone in it? 5 factorial?
- It's actually (5 - 1)! = 4! = 24. Why?
- Some arrangements are identical. For example, the arrangement:
    - A B C D E
- is the same as

    - D E A B C
- because, from everyone's frame of reference, everyone is still sitting in the same place. E.g. C is to the left of D in both of them, E is to the right of D in both of them etc.
- When you want to put the first person in, you have an option of 5 seats. But it doesn't matter what seat you put the first person in, the combination will still be the same, because from the first person's frame of reference, there's nobody around them. So the first person only has 1 combination of seat arrangement.
- If you have x seats, the number of ways you can arrange people is (x - 1)!

Combinations

- Let's say you had to find the number of ways to arrange the word GOOPPP.
- This arrangement: GOOPPP
- Is the same as this arrangement: GOOPPP
- So the answer is not 6!
- To find the answer, you need to divide 6! by the number of indistinguishable elements factorial multiplied together.
- For example, we have 2 indistinguishable O's and 3 indistinguishable P's in GOOPPP.
- Therefore, the number of arrangements is:
- $$\frac{6!}{2! \times 3!}$$

Counting subsets

- Let's say you have 10 different objects and you wanted to pick 4. What's the number of possible ways you can pick 4 objects out of 10 distinct objects?
- It is 10 choose 4, or 10C4, or $\frac{10!}{4!(10-4!)}$
- $$\binom{n}{k} := \frac{n!}{k!(n-k)!}$$

Binomial expansion

- $$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

Vandermonde Identity

- $$\sum_{k=0}^{t} \binom{n}{k} \binom{n-k}{t-k} = \binom{n}{t}$$

Multinomials

- $$\binom{n_1 + \cdots + n_K}{n_1, \cdots, n_K} \triangleq \frac{(n_1 + n_2 + \cdots + n_K)!}{n_1! n_2! \cdots n_K!}$$
- $$(a_1 + a_2 + a_3 + a_4)^{16} \quad = \quad a_1^{16} + \cdots + \binom{16}{6, 2, 5, 3} a_1^6 a_2^2 a_3^5 a_4^3 + \cdots + a_4^{16}$$
- Number of ways to arrange these blocks:
-
- Is this:
- $$\binom{16}{6, 2, 5, 3}$$
- So, in a way, this could be a shorthand for combinations

Probability and statistics

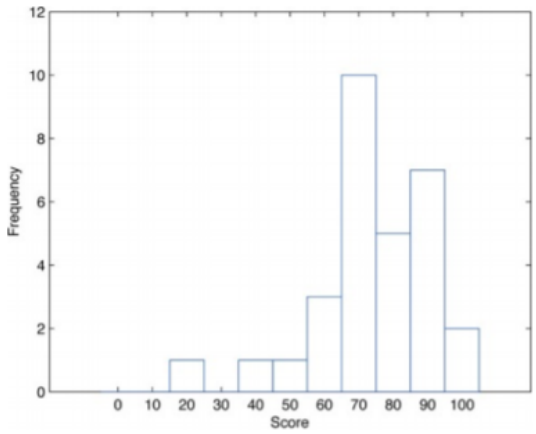
Descriptive statistics

Ordered stem and leaf diagram

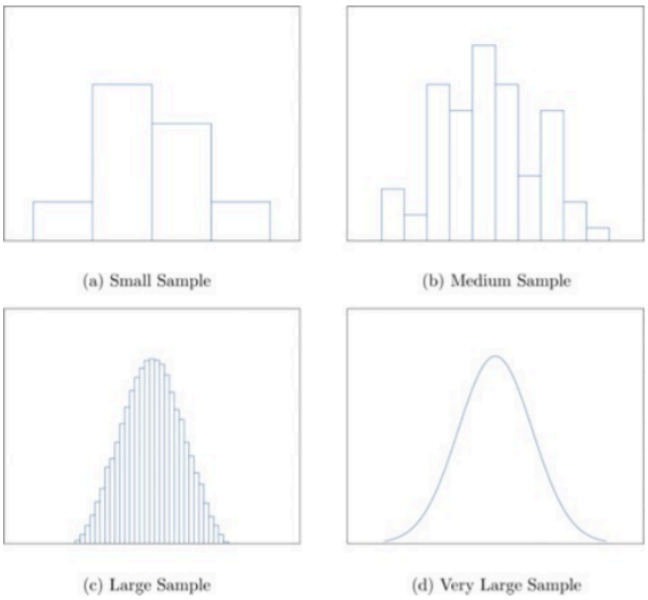
- 86 80 25 77 73 76 100 90 69 93
  90 83 70 73 73 70 90 83 71 95
  40 58 68 69 100 78 87 97 92 74
- Can be represented as
-

2	5
3	
4	0
5	8
6	8 9 9
7	0 0 1 3 3 3 4 6 7 8
8	0 3 3 6 7
9	0 0 0 2 3 5 7
10	0 0

Histograms

- The same set of data can also be represented as
- 

Sample sizes



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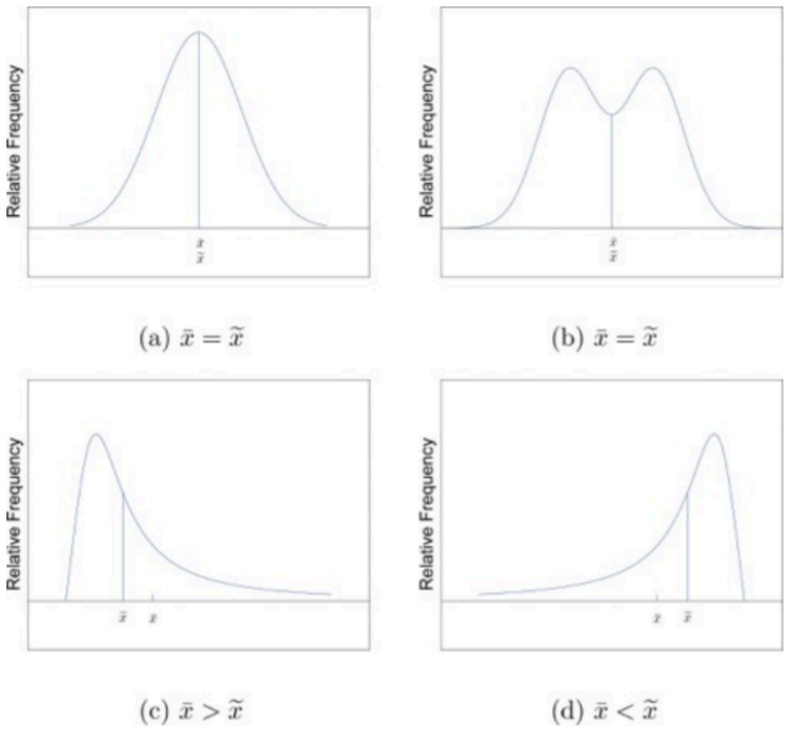
Measures of central location

- **Mean**: Add up all the data and divide it by how many there are (numerical average).
- **Median**: Sort the list and pick the one in the middle (centre element relative to magnitude).
- **Mode**: The most often data element (most frequent value).

Skewness of relative frequency histograms



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- Just so you know the notations:

$\bar{x}$	$\tilde{x}$
Mean	Median

Measures of variability

- **Range**: The highest data element MINUS the lowest data element (x_{max} - x_{min}).
- **Variance**: The average of the squared differences from the mean. It is the standard deviation squared.
  - $$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}$$
  - or
  - $$\sigma^2 = E[X^2] - (E[X])^2$$ - E[X] is the expected value function, defined as the "integral" under the probability distribution.
- **Standard deviation**: A measure of how spread out the numbers are. It is the square root of the variance.
  - $$\sigma = \sqrt{\left(\frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}\right)}$$
- Variance > Standard deviation for variance > 1.

Relative position of data

Percentiles

- The percentile P of a data value x is the percent of how high it is compared to the other data values. with 100% being the highest and 0% being the lowest.
- Formula: P = $\frac{number\ of\ elements\ less\ than\ or\ equal\ to\ x}{number\ of\ elements\ in\ total} \times 100$
- Example:
  - You have this data set:
  - 2, 6, 8, 10, 12, 18, 22
  - What is the percentile of '8'?
  - There are 3 values less than or equal to 8.
  - 2, 6, 8, 10, 12, 18, 22
  - There are 7 values in total.
  - Therefore P = $\frac{3}{7} \times 100$= 42%

Quartiles

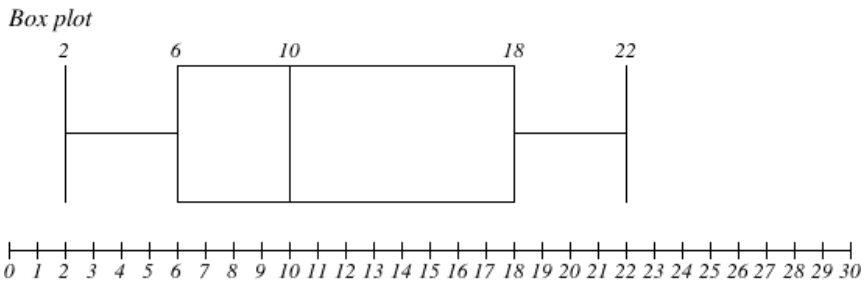
- Quartiles are like percentiles, but they split the data into quarters.
- The first quartile Q_1 is the median of the lower half of the set

- The second quartile Q_2 is the median of the whole set
- The third quartile Q_3 is the median of the upper half of the set
- Suppose you have n ordered elements,
    - To find the median (Q_2): **(n+1)/2** th value $\frac{(n+1)}{2}$
    - To find the lower quartile (Q_1): **(n+1)/4** th value $\frac{(n+1)}{4}$
    - To find the upper quartile (Q_3): **3(n+1)/4** th value $\frac{3(n+1)}{4}$
- Example:
    - You have this data set:
    - 2, 6, 8, 10, 12, 18, 22
    - What are the three quartiles?
    - 2, 6, 8, 10, 12, 18, 22
    - Q_2 = 10, because that's the median
    - The lower half is now 2, 6, 8, so Q_1 = 6
    - The upper half is now 12, 18, 22, so Q_3 = 18

Box plot

- A box plot is simply a visual representation of quartiles.

-
- The lowest value in the set
- The first quartile (median of lower half)
- The second quartile (median of the whole set)
- The third quartile (median of upper half)
- The largest value in the set
- Example:
    - You have this data set:
    - 2, 6, 8, 10, 12, 18, 22



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Interquartile range

- The interquartile range is the third quartile minus the first quartile:
- Q_3 - Q_1

The Empirical Rule

- If a data set has an approximately bell-shaped relative frequency histogram, then:
    - approximately 68% of the data lie within one standard deviation of the mean.
    - approximately 95% of the data lie within two standard deviations of the mean.
    - approximately 99.7% of the data lies within three standard deviations of the mean.
- Example:
    - The standard deviation for this set:
    - 2, 6, 8, 10, 12, 18, 22
    - is 6.4. The mean is 11.1.
    - One standard deviation from the mean is:
    - 11.1 ± 6.4
    - Which can be between
    - 4.7 < x < 17.5
    - According to the empirical rule, approximately 68% of the data values in the set are between 4.7 and 17.5.
    - If my range was spread out to two standard deviations:
    - $11.1 \pm 2(6.4)$
    - 11.1 ± 12.8
    - -1.7 < x < 23.9
    - According to the empirical rule, approximately 95% of the data values in the set are between -1.7 and 23.9.

Chebyshev's Theorem

- For any numerical data set
- at least $\frac{3}{4}$ of the data lie within two standard deviations of the mean
- at least $\frac{8}{9}$ of the data lie within three standard deviations of the mean
- at least $1 - \frac{1}{k^2}$ of the data lie within k standard deviations of the mean, where k is any positive whole number that is greater than 1.
- This is different from the empirical rule from the use of the words 'AT LEAST'. According to this theorem, it would be impossible for **only** ½ of the data to lie within two standard deviations of the mean; it has to be at least ¾.

Basic concepts of probability

Random events

- A random event is an event whose frequency will tend towards a stable value.
- Example:
    - You flip a coin 10 times. You get 3 heads and 7 tails.
    - You flip a coin 1000 times. You get 519 heads and 481 tails.

- If you flipped the coin infinite times, you'd see that half the time you'd get heads, and half the time you'd get tails. This shows that flipping heads or tails are random events.

Sample space

- A sample space is the set of all possible combinations.
- Example:
  - Sample space of flipping a coin:
  - S = { H, T }
  - Sample space of rolling a dice:
  - S = { 1, 2, 3, 4, 5, 6 }

Events

- An event E is a subset of sample space S.
- Example:
  - Sample space of rolling a dice:
  - S = { 1, 2, 3, 4, 5, 6 }
  - Event that you will roll an even number:
  - E = { 2, 4, 6 }
  - As you can see, event E is a subset of sample space S.
- If event E1 is true or event E2 is true, then that event can be represented as:
  - $E_1 \cup E_2$
- If event E1 is true and event E2 is true, then that event can be represented as:
  - $E_1 \cap E_2$
  - Or just
  - $E_1 E_2$
- The empty set is also an event. This event is impossible; it can never happen.
  - Ø
- When $E_1 \cap E_2 = \emptyset$, then those two events are mutually exclusive.
- That means that they both cannot be true at the same time; only one must be true, or neither of them.
- The complement of E, E^c, is true when E does not occur. Think of it like the 'inverse'.

Probability of events

- The probability of an event E can be expressed like this:
  - P(E)
- The probability of an event must satisfy these axioms:
  - P(E) >= 0 where E is any event
  - P(S) = 1 where S is a sample space
  - if two events are mutually exclusive:
    - P($E \cup F$) = P(E) + P(F)
  - if two events are NOT mutually exclusive:
    - P($E \cup F$) = P(E) + P(F) - P(EF)

- More rules:
  - $(E \cup F)^c = E^c F^c$
  - $E\emptyset = \emptyset$
  - $E \cup \emptyset = E$
  - $F = FE \cup FE^c$
  - $E \cup F = F \cup EF^c = E \cup FE^c$
  - $(EF)^c = E^c \cup F^c$
- The probability of an event is:
  - $E = \{ D_1, D_2, D_3, ... \}$
  - $P(E) = P(D_1) + P(D_2) + P(D_3) + ...$
- Another probability of an event is:
  - $\frac{size\ of\ event\ E}{size\ of\ sample\ space\ S}$

Tuple

- r-tuple = $(n_1, n_2, n_3 ... n_r)$
- Order matters

Combinatorial analysis

- Let A be a set of r-tuples, $\{ x_1\ x_2 ... x_r \}$, such that there are:
- Firstly: n_1 different ways in which to choose x_1,
- Secondly: n_2 different ways in which to choose x_2, ...
- Finally: n_r different ways in which to choose x_r.
- Then A contains $N(A) = n_1 * n_2 * ... * n_r$ different r-tuples.
- Example:
  - You have a dice and you roll it 3 times.
  - Each iteration, a 3-tuple can be made of what you just did.
  - E.g. if you rolled a 2, then a 5, then a 6, the tuple to represent that could be (2, 5, 6)
  - You cannot roll the same value as you have done before, e.g. (2, 4, 6) is allowed but (2, 3, 2) is not.
  - The set A contains all the possible 3-tuples that you can do when you roll the dice 3 times.
  - In general, the 3-tuples look like this:
  - (x_1, x_2, x_3)
  - The number of ways to choose x_1 is 6.
  - The number of ways to choose x_2 is 5.
  - The number of ways to choose x_3 is 4.
  - Therefore, the size of set A is 6 * 5 * 4 which is 120. There are 120 different possible outcomes of this.

Basic sampling

- Number of ways to draw n balls from an urn containing M distinguishable balls
  - without replacement: M choose n

- with replacement: M^n

Subset of sample

- Set S = {1, 2, ... , N }
- How many subsets of S of size k?
- Answer: N choose k

Partitions of a set of size k

- The number of ways you can partition a set of size N so that the 1st subset is size k_1, 2nd subset is size k_2 etc. and $k_1 + k_2 + ... + k_r = N$
- Answer: (N choose k_1) * ((N - k_1) choose k_2) - ... - ((N - k_1 - k_2 - ... - k_r) choose k_r)

Number of successes in a sample

- An urn contains M balls of which M_W are white and $M_R = M - M_W$ are red. A sample of size n is drawn either with or without replacement. Let k be an integer between 0 and n. What is the probability that the sample will contain exactly k white balls?
- Sampling without replacement:
  - $$P(A_k) = \binom{n}{k} \frac{(M_w)_k (M - M_w)_{n-k}}{(M)_n}$$
- Sampling with replacement:
  - $$P(A_k) = \binom{n}{k} \frac{(M_w)^k (M - M_w)^{n-k}}{(M)^n}$$
- Notation:
  - $$(M)_n = \frac{M!}{(M-n)!}$$
- Dumb it down for me:
  - An urn contains 10 balls. 4 are white, and 6 are red. You take out 5. What's the probability you will get 3 white balls?
  - n choose k is 5 choose 3, which is 10, therefore there are 10 possible ways of picking out white balls.
  - Without replacement:
    - The first time you pick out a white ball, there are 4 different white balls to pick.
    - The second time you pick out a white ball, there are 3 different white balls to pick, etc.
    - You pick out 3 white balls and 2 red balls, so the chance of that happening through one picking is:
    - $\frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{6}{7} \times \frac{5}{6} = \frac{1}{42}$

    - But you could also pick out 2 white balls, a red ball, then a white ball, then another red ball. The chance of that would be:
    - $\frac{4}{10} \times \frac{3}{9} \times \frac{6}{8} \times \frac{2}{7} \times \frac{5}{6} = \frac{1}{42}$
    - We need to cover every combination of picking balls, so we multiply n choose k (10) to that chance, giving us $\frac{5}{21}$which is around 24%.
- With replacement:
    - Each time you pick out a white ball, there are 4 different white balls to pick.
    - Each time you pick out a red ball, there are 6 different balls to pick.
    - You pick out 3 white balls and 2 red balls, so the chance of that happening through one picking is:
    - $\frac{4}{10} \times \frac{4}{10} \times \frac{4}{10} \times \frac{6}{10} \times \frac{6}{10} = \frac{72}{3125}$
    - But you could also pick out 2 white balls, a red ball, then a white ball, then another red ball. The chance of that would be:
    - $\frac{4}{10} \times \frac{4}{10} \times \frac{6}{10} \times \frac{4}{10} \times \frac{6}{10} = \frac{72}{3125}$
    - We need to cover every combination of picking balls, so we multiply n choose k (10) to that chance, giving us $\frac{144}{625}$which is around 23%.

Difference between k successes and successes on k specified draws

- Sample size 3, without replacement, urn with 4 white and 2 red
- Probability of first being white, second being white and third being red
    - $\frac{4}{6} \times \frac{3}{5} \times \frac{2}{4}$or
    - $\frac{(4C2)(2C1)}{(6C3)}$

Conditional probability

- P(B|A) = probability of B given that A occurs
- P(B|A) = $\frac{N_{AB}}{N_A}$
- P(B|A) = $\frac{P(AB)}{P(A)}$ if P(A) > 0

Basic probability theory

- B is independent of A given P(B|A) = P(B)
- A and B are independent if P(BA) = P(B)P(A)
- A, B, C are independent if
- P(AB) = P(A)P(B)
- P(AC) = P(A)P(C)
- P(BC) = P(B)P(C)
- P(ABC) = P(A)P(B)P(C)

Total probability

- A is an event in sample space S. Let B_1, B_2, B_3 ... Bn be the partitions of A.
- $P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + ... + P(B_n)P(A|B_n)$

Bayes' Theorem

- A is a possible event. Let B_1, B_2, B_3 ... Bn be the partitions of A.
- $P(B_i|A) = \frac{P(B_i)P(A|B_i)}{P(A)}$

Distributions

Random variables

- A random variable X is a variable whose possible values are numerical outcomes of a random phenomenon, like the value of a dice when it's rolled. There are two types of variable: discrete and continuous.

Discrete random variables

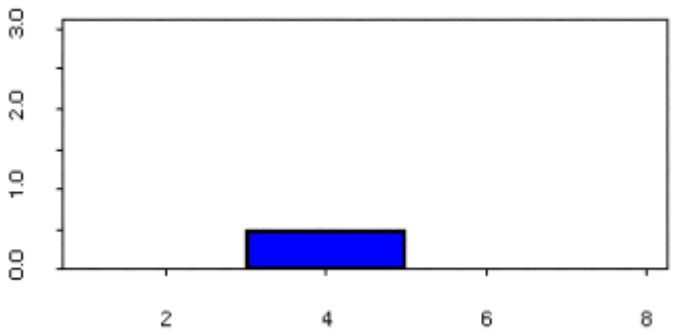
- A discrete random variable is a variable which may take on the form of an integer such as 0, 1, 2, 3, 4 ...
- Discrete random variables are usually counts.
- The probability of each outcome MUST:
  - Sum up to 1
  - Each must be between 0 and 1
- Examples:
  - Value of a dice roll
  - Number of children in a family
  - Number of defective light bulbs in a box of ten
- The probabilities of all possible discrete random variables can be expressed in a table of outcomes to probabilities. This is called a **probability mass function**.
- Example:

Outcome	1	2	3	4	5	6
Probability	1/6	1/6	1/6	1/6	1/6	1/6

- pmf(5) = ⅙
- where pmf = probability mass function
- Expected value (mean) of discrete data:
- $\mathbb{E}[X] = \mu = \sum_{x \in S} x f_X(x) = \sum_{i=1}^{m} x_i p_i$
- Variance of discrete data:

- $$\sigma^2 = \mathbb{E}[(X-\mu)^2] = \sum_{x \in S}(x-\mu)^2 f_X(x) = \sum_{i=1}^{m}(x_i-\mu)^2 p_i$$

Continuous random variables

- A continuous random variable is a variable which can take an infinite number of values. They are usually expressed not by integers, but by a range, like 1 to 10.
- They are in the set **R** (real numbers).
- These variables are usually measurements.
- Examples:
  - Height / Weight of a person
  - Amount of sugar in an orange
  - Time required to run a mile
- The probability of a continuous random variable is represented by the area under a curve (or integral, in maths terms) which is called the variable's **probability density function**.
- The curve, which represent a function p(x), MUST:
  - Have no negative values: p(x) >= 0
  - Have a total area under the curve of 1
- Example of a continuous random variable's probability density function:
  - 

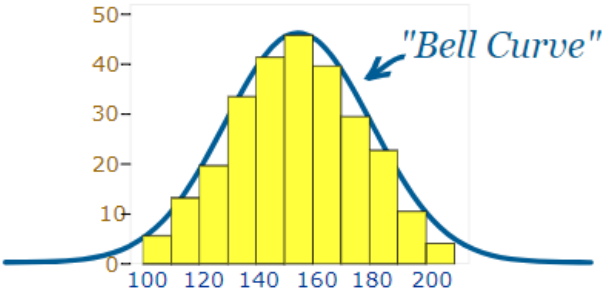
  - In this example, a variable X can be between 3 and 5. It could be 3.56, it could be 5, it could even be 4.10582905310346451609371340934 6. It just keeps on going, which is why a continuous random variable can hold an infinite amount of values.
  - To find the probability that X will be between 3 and 4, P(3 < x < 4), you must find the area under the 'curve' between the values x = 3 and x = 4.
  - When you find that area, it should be 0.5. This means that P(3 < x < 4) = 0.5.
- Expected value (mean) of continuous data:
  - $$\mu = \mathbb{E}[x] = \int_{-\infty}^{\infty} x f(x) dx$$
- Variance of continuous data:
  - $$\sigma^2 = \mathbb{E}[(X-\mu)^2] = \int_{-\infty}^{\infty}(x-\mu)^2 f(x) dx$$

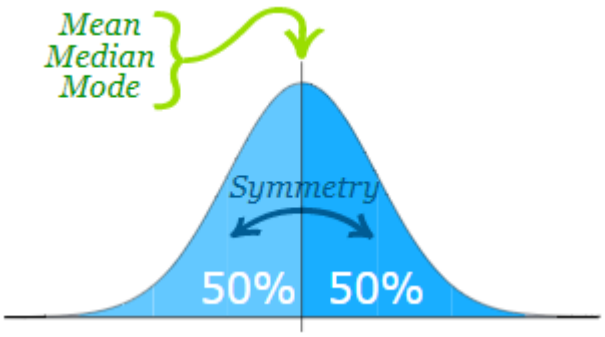
Binomial Distribution

- Bernoulli trials are independent repeated random trials that only have two outcomes: success and failure.
- If the probability of success is p, then the probability of failure ‘q’, is q = 1 - p.
- A binomial distribution is a series of Bernoulli trials.
- Binomial distributions will only apply to **discrete random variables**.
- B(n, p) denotes a series of Bernoulli trials (a binomial distribution) with ‘n’ trials and a probability of success ‘p’. Usually, we are interested in the number of successes and not the order in which they occur.
- Example:
  - You roll a dice 10 times. You decide that rolling a ‘6’ is a success, and everything else is a failure.
  - To represent this as a binomial distribution, you could write B(10, ⅙).
- Probability of exactly k successes in a binomial distribution B(n, p) is
  - $$P[k] = \binom{n}{k} p^k q^{n-k}$$
- Mean (expected number of successes) of a binomial distribution:
  - $$\mu = np$$
- Variance of a binomial distribution:
  - $$\sigma^2 = npq$$
- Probability of at least k successes is:
  - $$P[x >= k] = P[k] + P[k+1] + P[k+2] + ... + P[n]$$
  - Or, you can go the other way round, if it’s shorter:
    - $$\begin{aligned} & P[x >= k] \\ &= 1 - P[x < k] \\ &= P[0] + P[1] + P[2] + ... + P[k-1] \end{aligned}$$
- Example:
  - You roll a dice 10 times. You decide that rolling a ‘6’ is a success, and everything else is a failure.
  - Probability of exactly 4 successes:
    - $$\binom{10}{4} (\frac{1}{6})^4 (\frac{5}{6})^{10-4} = 0.0543$$

- Mean (expected number of successes):
- $$\mu = (10)(\frac{1}{6}) = \frac{5}{3} = 1.67$$
- Probability of at least 2 successes:
- $$\begin{aligned} & P(x >= 2) \\ &= 1 - P(x < 2) \\ &= 1 - (P(0) + P(1)) \\ &= 1 - ((\frac{5}{6})^{10} + \binom{10}{1}(\frac{1}{6})^1(\frac{5}{6})^9) \\ &= 1 - 0.484516... \\ &= 0.515483... \end{aligned}$$

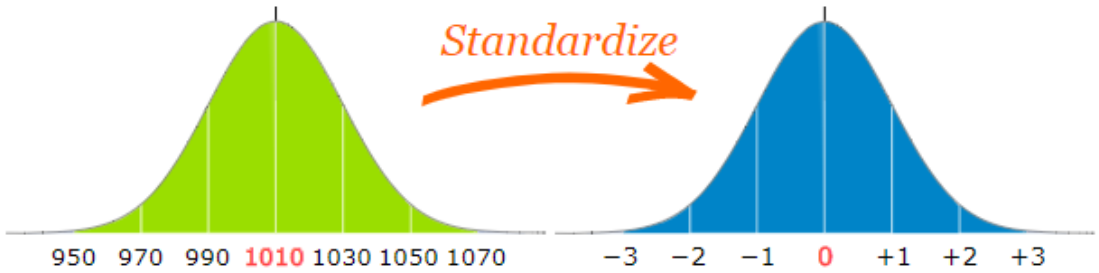
Normal Distribution

- A normal distribution, or a Gaussian distribution, is an example of a continuous random variable whose density is a bell-shaped graph.
- 

A Normal Distribution
- A normal distribution has these following requirements:
  - mean = median = mode
  - symmetry around the center
  - 50% of the values less than the mean and 50% greater than the mean
- 

- Like how a binomial distribution is defined by the number of trials n and the probability of success k, a normal distribution is defined by the mean μ and the variance σ^2.
- Normal distributions can be represented like so: $N(\mu, \sigma^2)$
- With μ and σ^2, the corresponding density function is:
- $$f(x) = \frac{1}{\sigma\sqrt{2\pi}} exp[-\frac{1}{2}[\frac{x-\mu}{\sigma}]^2], \quad -\infty < x < \infty$$
- You probably won't have to remember this.

Standard Normal Distribution

- You can convert a normal distribution into a *standard* normal distribution.
- The difference is that each of the standard normal distribution's units are actually multiples of the standard deviation, making it easier to do calculations and compare variables with each other.
- 

- To convert a random variable (a unit in the normal distribution, like 1030) into a standard random variable (a unit in the standard normal distribution, like +1) you can use this formula:
  - $$Z = \frac{X-\mu}{\sigma}$$
  - Where Z is the standard random variable,
  - X is the random variable you want to convert,
  - μ is the mean, and
  - σ is the standard deviation.

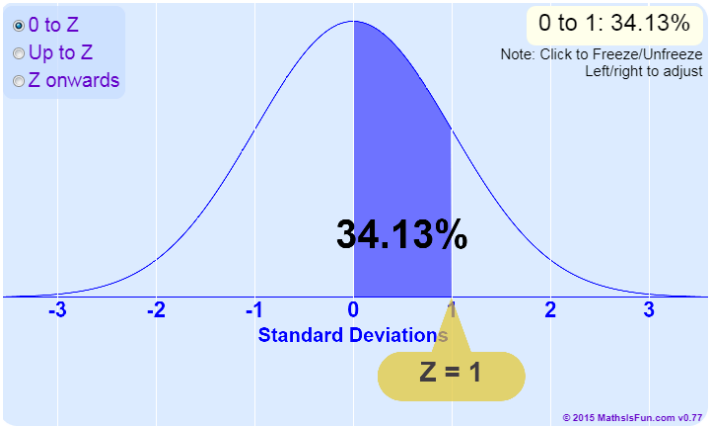
Probabilities for standard normal distribution

- Ones that you should just know (or can just figure out):
  - $P[Z <= 0] = 0.5$
  - $P[Z >= 0] = 0.5$
  - $P[Z <= x] = P[Z >= -x]$
  - $P[Z >= x] = P[Z <= -x]$
  - $P[x_1 >= Z >= x_2] = P[-x_2 >= Z >= -x_1]$
  - $P[x_1 <= Z <= x_2] = P[Z <= x_2] - P[Z <= x_1]$
- You can find the probability of a variable being in a certain range.
- Using a standard normal distribution table, you can look up the probability that a standard random variable will lie between 0 and a certain value:

-

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772

- For example, you could look up P[0 <= Z <= 0.43] by looking up 0.4 and 0.03 (since 0.43 = 0.4 + 0.03):
-

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772

- Therefore P[0 <= Z <= 0.43] = 0.1664
- The full table can be found here: Standard Normal Distribution Table
- Examples:
  - When you have other probability problems, split it up into things that are either obvious or things you can look up in a table:
  - $P[x_1 <= Z <= x_2]$ =
    - $P[0 <= Z <= x_2] - P[0 <= Z <= x_1]$
  - P[Z >= x] =
    - 1 - (P[Z <= 0] + P[0 <= Z <= x])
    - 1 - (0.5 + P[0 <= Z <= x])
  - P[Z <= x] =
    - P[Z <= 0] + P[0 <= Z <= x]
    - 0.5 + P[0 <= Z <= x]
- While solving probability problems like this, visualise a standard normal distribution graph in your head (or draw it out as a sketch), like this:
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